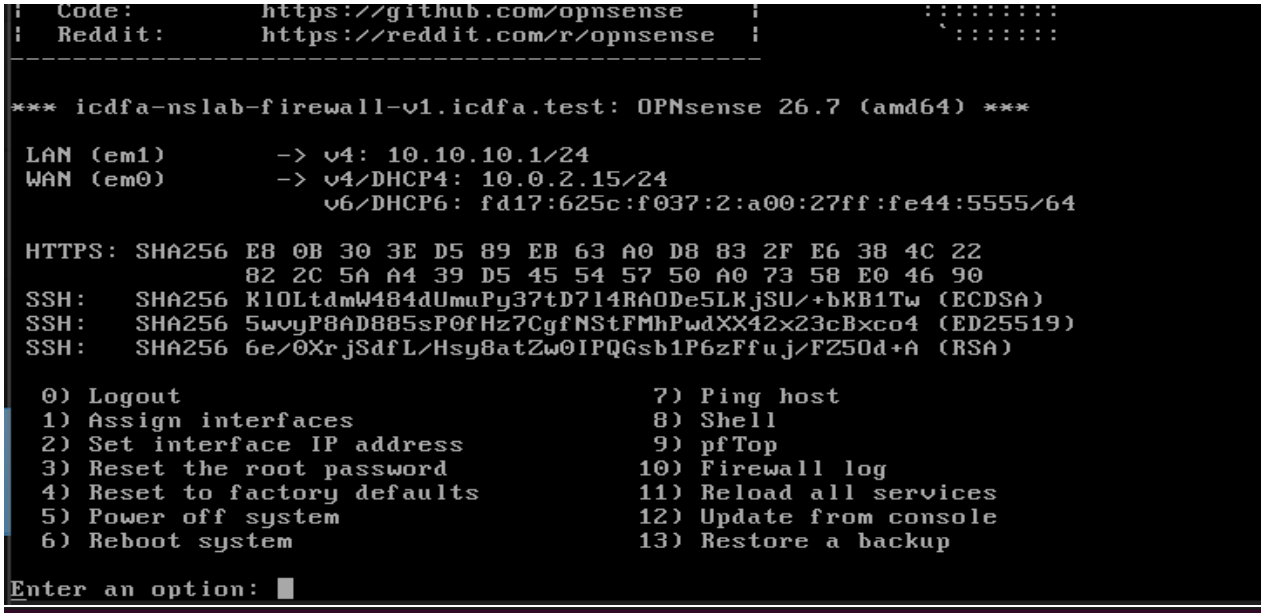
FULL NAME	USMAN IBRAHIM
REGISTRATION NUMBER	C11/26/FCDF/17159

LABBORATORY REPORT ON 2 VM NETWORK COMMISIONING AND CONNECTIVITY VERIFICATION






```
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br address
lo               UNKNOWN        127.0.0.1/8
enp0s17          UP             10.10.10.163/24
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s17 proto dhcp src 10.10.10.163 metric 20100
10.10.10.0/24 dev enp0s17 proto kernel scope link src 10.10.10.163 metric 100
icdfa@icdfa-nslab-client-v1:~$ resolvectl status
Global
         Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
  resolv.conf mode: stub

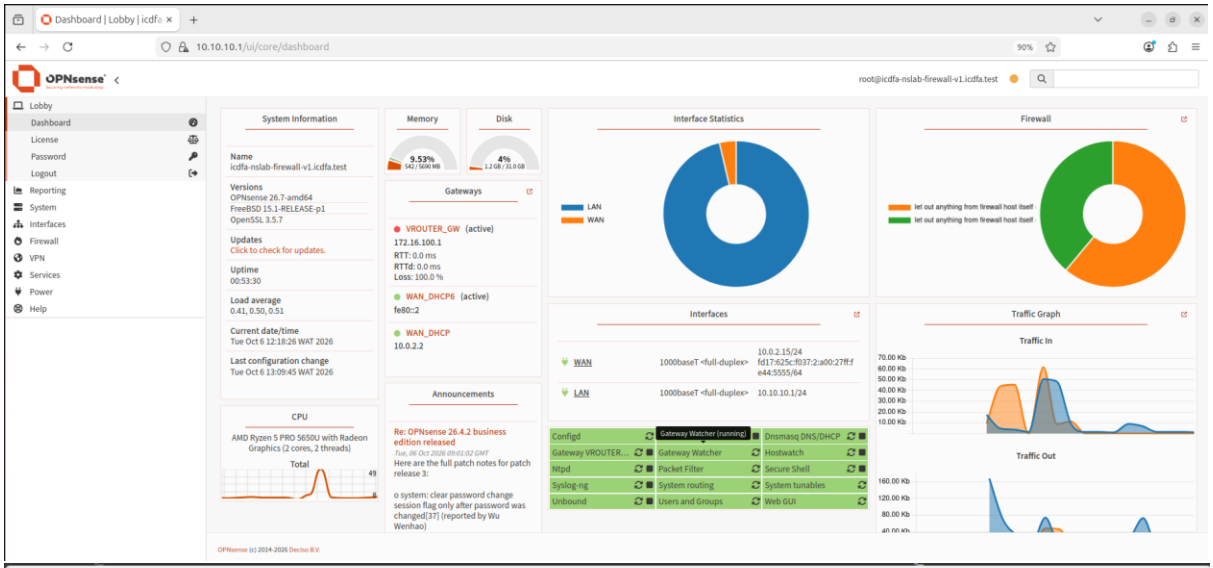
Link 2 (enp0s17)
    Current Scopes: DNS
         Protocols: +DefaultRoute -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
Current DNS Server: 10.10.10.1
       DNS Servers: 10.10.10.1
        DNS Domain: icdfa.test
icdfa@icdfa-nslab-client-v1:~$
```

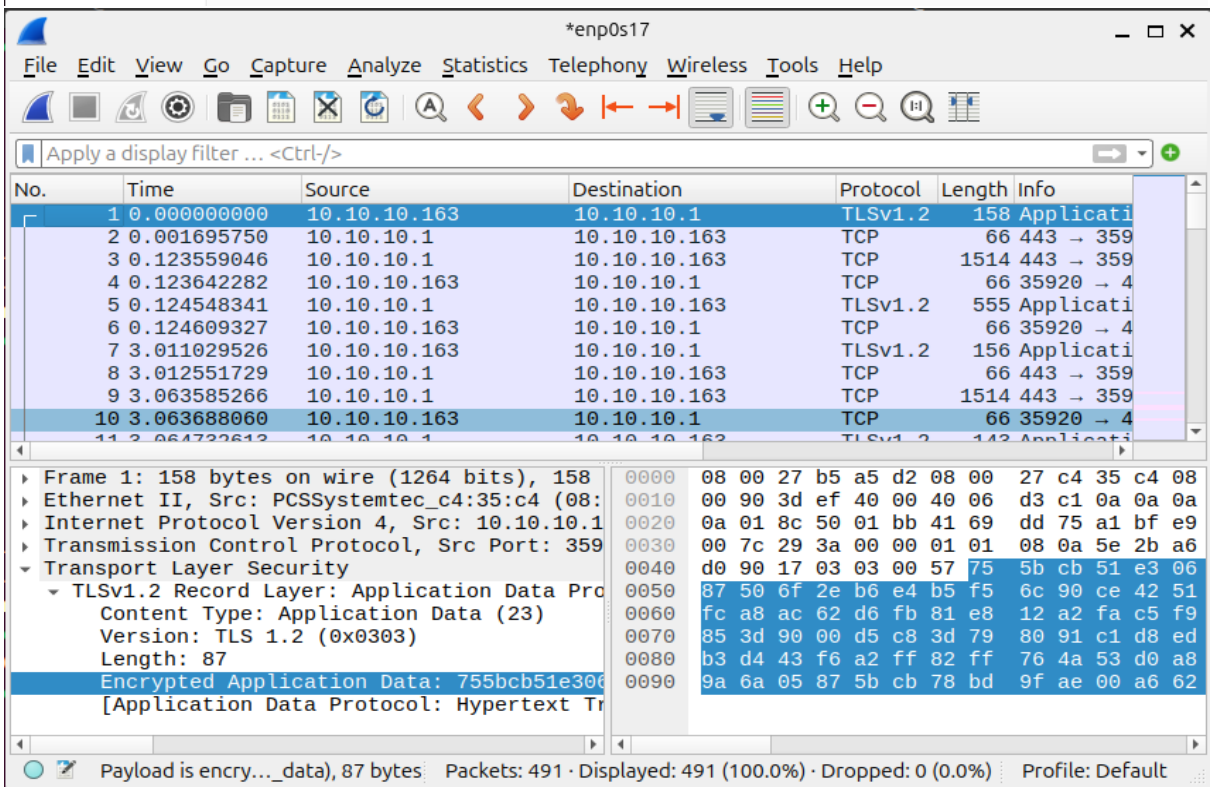
```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 10.10.10.1
PING 10.10.10.1 (10.10.10.1) 56(84) bytes of data.
64 bytes from 10.10.10.1: icmp_seq=1 ttl=64 time=4.77 ms
64 bytes from 10.10.10.1: icmp_seq=2 ttl=64 time=1.72 ms
64 bytes from 10.10.10.1: icmp_seq=3 ttl=64 time=1.52 ms
64 bytes from 10.10.10.1: icmp_seq=4 ttl=64 time=1.71 ms

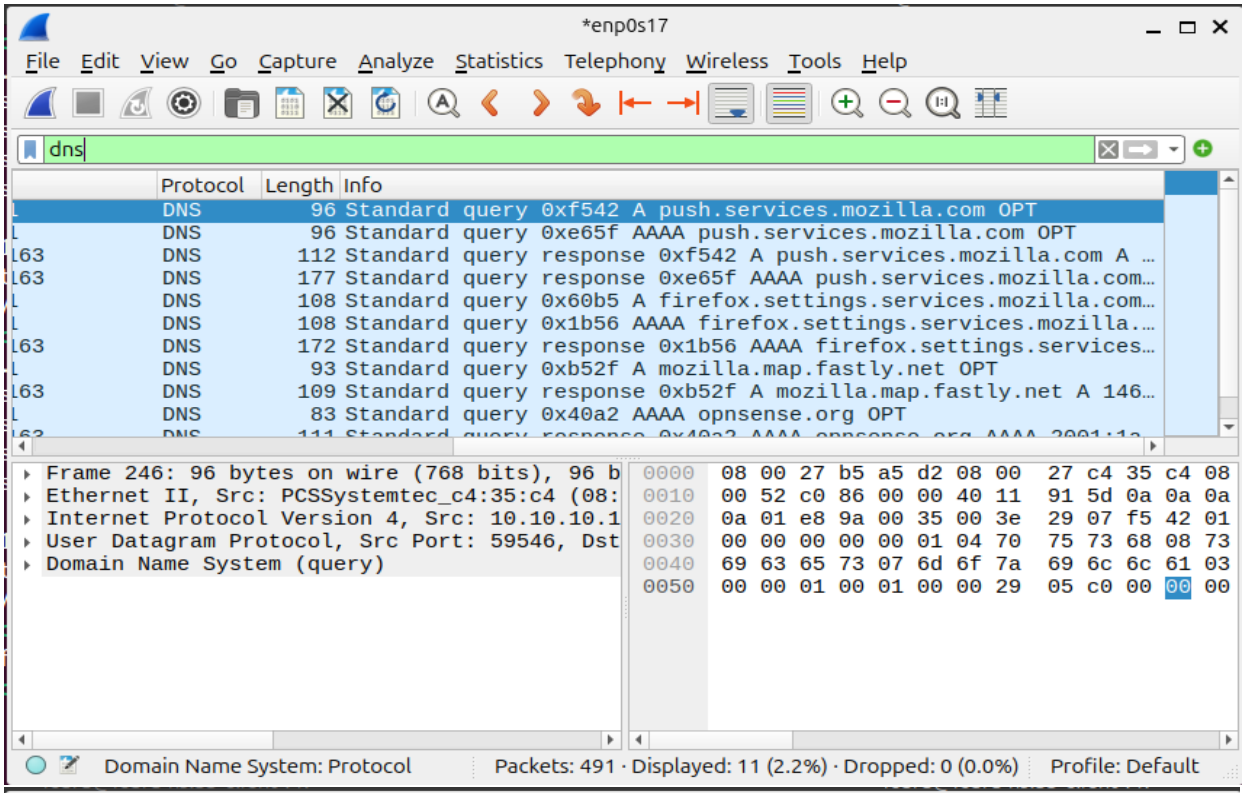
--- 10.10.10.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 1.518/2.428/4.767/1.352 ms
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=62 time=199 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=62 time=199 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=62 time=214 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=62 time=195 ms

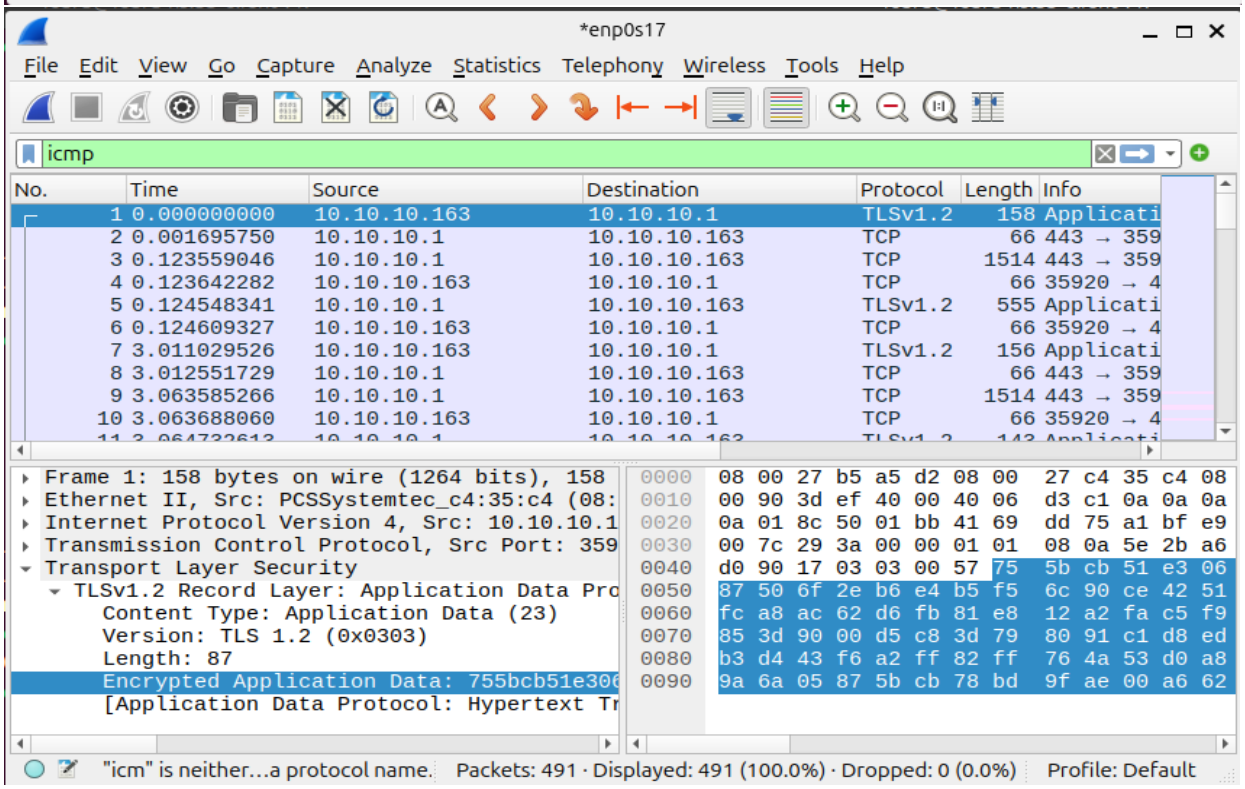
--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 195.282/201.766/213.791/7.106 ms
icdfa@icdfa-nslab-client-v1:~$ getent hosts opnsense.org
2001:1af8:2050:a001:1::1 opnsense.org
icdfa@icdfa-nslab-client-v1:~$ curl -I https://opnsense.org
HTTP/1.1 200 OK
Date: Tue, 06 Oct 2026 11:16:54 GMT
Server: Apache
Link: <https://opnsense.org/wp-json/>; rel="https://api.w.org/", <https://opnsense.org/
wp-json/wp/v2/pages/199>; rel="alternate"; title="JSON"; type="application/json", <http
s://opnsense.org/>; rel=shortlink
Strict-Transport-Security: max-age=15552000; includeSubDomains; preload
Content-Type: text/html; charset=UTF-8

icdfa@icdfa-nslab-client-v1:~$
```


Dashboard | Lobby | icdfa
10.10.10.1/ui/core/dashboard
OPNsense
root@icdfa-nslab-firewall-v1.icdfa.test
Lobby
Dashboard
License
Password
Logout
Reporting
System
Interfaces
Firewall
VPN
Services
Power
Help
System Information
Name
icdfa-nslab-firewall-v1.icdfa.test
Versions
OPNsense 26.7-amd64
FreeBSD 15.1-RELEASE-p1
OpenSSL 3.5.7
Updates
Click to check for updates.
Uptime
00:53:30
Load average
0.41, 0.50, 0.51
Current date/time
Tue Oct 6 12:18:26 WAT 2026
Last configuration change
Tue Oct 6 13:09:45 WAT 2026
Memory
9.53%
Disk
4%
Gateways
VROUTER_GW (active)
172.16.100.1
RTT: 0.0 ms
RTTd: 0.0 ms
Loss: 100.0 %
WAN_DHCP6 (active)
fe80::2
WAN_DHCP
10.0.2.2
Announcements
Re: OPNsense 26.4.2 business edition released
Here are the full patch notes for patch release 3:
o system: clear password change session flag only after password was changed[37] (reported by Wu Wenhao)
CPU
AMD Ryzen 5 PRO 5650U with Radeon Graphics (2 cores, 2 threads)
Total
Interface Statistics
LAN
WAN
Firewall
Interfaces
WAN
1000baseT <full-duplex>
10.0.2.15/24
fd17:625c:f037:2:a00:27ff:fe44:5555/64
LAN
1000baseT <full-duplex>
10.10.10.1/24
Configd
Gateway Watcher (running)
Dnsmasq DNS/DHCP
Gateway VROUTER...
Gateway Watcher
Hostwatch
Ntpd
Packet Filter
Secure Shell
Syslog-ng
System routing
System tunables
Unbound
Users and Groups
Web GUI
Traffic Graph
Traffic In
Traffic Out


*enp0s17
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000000 10.10.10.163 10.10.10.1 TLSv1.2 158 Applicati
2 0.001695750 10.10.10.1 10.10.10.163 TCP 66 443 → 359
3 0.123559046 10.10.10.1 10.10.10.163 TCP 1514 443 → 359
4 0.123642282 10.10.10.163 10.10.10.1 TCP 66 35920 → 4
5 0.124548341 10.10.10.1 10.10.10.163 TLSv1.2 555 Applicati
6 0.124609327 10.10.10.163 10.10.10.1 TCP 66 35920 → 4
7 3.011029526 10.10.10.163 10.10.10.1 TLSv1.2 156 Applicati
8 3.012551729 10.10.10.1 10.10.10.163 TCP 66 443 → 359
9 3.063585266 10.10.10.1 10.10.10.163 TCP 1514 443 → 359
10 3.063688060 10.10.10.163 10.10.10.1 TCP 66 35920 → 4
Frame 1: 158 bytes on wire (1264 bits), 158
Ethernet II, Src: PCSSystemtec_c4:35:c4 (08:
Internet Protocol Version 4, Src: 10.10.10.1
Transmission Control Protocol, Src Port: 359
Transport Layer Security
TLSv1.2 Record Layer: Application Data Pro
Content Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 87
Encrypted Application Data: 755bcb51e306
[Application Data Protocol: Hypertext Tr
0000 08 00 27 b5 a5 d2 08 00 27 c4 35 c4 08
0010 00 90 3d ef 40 00 40 06 d3 c1 0a 0a 0a
0020 0a 01 8c 50 01 bb 41 69 dd 75 a1 bf e9
0030 00 7c 29 3a 00 00 01 01 08 0a 5e 2b a6
0040 d0 90 17 03 03 00 57 75 5b cb 51 e3 06
0050 87 50 6f 2e b6 e4 b5 f5 6c 90 ce 42 51
0060 fc a8 ac 62 d6 fb 81 e8 12 a2 fa c5 f9
0070 85 3d 90 00 d5 c8 3d 79 80 91 c1 d8 ed
0080 b3 d4 43 f6 a2 ff 82 ff 76 4a 53 d0 a8
0090 9a 6a 05 87 5b cb 78 bd 9f ae 00 a6 62
Payload is encry..._data), 87 bytes
Packets: 491 · Displayed: 491 (100.0%) · Dropped: 0 (0.0%)
Profile: Default


*enp0s17
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
Protocol Length Info
DNS 96 Standard query 0xf542 A push.services.mozilla.com OPT
DNS 96 Standard query 0xe65f AAAA push.services.mozilla.com OPT
DNS 112 Standard query response 0xf542 A push.services.mozilla.com A …
DNS 177 Standard query response 0xe65f AAAA push.services.mozilla.com…
DNS 108 Standard query 0x60b5 A firefox.settings.services.mozilla.com…
DNS 108 Standard query 0x1b56 AAAA firefox.settings.services.mozilla.…
DNS 172 Standard query response 0x1b56 AAAA firefox.settings.services…
DNS 93 Standard query 0xb52f A mozilla.map.fastly.net OPT
DNS 109 Standard query response 0xb52f A mozilla.map.fastly.net A 146…
DNS 83 Standard query 0x40a2 AAAA opnsense.org OPT
Frame 246: 96 bytes on wire (768 bits), 96 b
Ethernet II, Src: PCSSystemtec_c4:35:c4 (08:
Internet Protocol Version 4, Src: 10.10.10.1
User Datagram Protocol, Src Port: 59546, Dst
Domain Name System (query)
0000 08 00 27 b5 a5 d2 08 00 27 c4 35 c4 08
0010 00 52 c0 86 00 00 40 11 91 5d 0a 0a 0a
0020 0a 01 e8 9a 00 35 00 3e 29 07 f5 42 01
0030 00 00 00 00 00 01 04 70 75 73 68 08 73
0040 69 63 65 73 07 6d 6f 7a 69 6c 6c 61 03
0050 00 00 01 00 01 00 00 29 05 c0 00 00 00
Domain Name System: Protocol
Packets: 491 · Displayed: 11 (2.2%) · Dropped: 0 (0.0%)
Profile: Default


*enp0s17
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
icmp
No. Time Source Destination Protocol Length Info
1 0.000000000 10.10.10.163 10.10.10.1 TLSv1.2 158 Applicati
2 0.001695750 10.10.10.1 10.10.10.163 TCP 66 443 → 359
3 0.123559046 10.10.10.1 10.10.10.163 TCP 1514 443 → 359
4 0.123642282 10.10.10.163 10.10.10.1 TCP 66 35920 → 4
5 0.124548341 10.10.10.1 10.10.10.163 TLSv1.2 555 Applicati
6 0.124609327 10.10.10.163 10.10.10.1 TCP 66 35920 → 4
7 3.011029526 10.10.10.163 10.10.10.1 TLSv1.2 156 Applicati
8 3.012551729 10.10.10.1 10.10.10.163 TCP 66 443 → 359
9 3.063585266 10.10.10.1 10.10.10.163 TCP 1514 443 → 359
10 3.063688060 10.10.10.163 10.10.10.1 TCP 66 35920 → 4
Frame 1: 158 bytes on wire (1264 bits), 158
Ethernet II, Src: PCSSystemtec_c4:35:c4 (08:
Internet Protocol Version 4, Src: 10.10.10.1
Transmission Control Protocol, Src Port: 359
Transport Layer Security
TLSv1.2 Record Layer: Application Data Pro
Content Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 87
Encrypted Application Data: 755bcb51e306
[Application Data Protocol: Hypertext Tr
0000 08 00 27 b5 a5 d2 08 00 27 c4 35 c4 08
0010 00 90 3d ef 40 00 40 06 d3 c1 0a 0a 0a
0020 0a 01 8c 50 01 bb 41 69 dd 75 a1 bf e9
0030 00 7c 29 3a 00 00 01 01 08 0a 5e 2b a6
0040 d0 90 17 03 03 00 57 75 5b cb 51 e3 06
0050 87 50 6f 2e b6 e4 b5 f5 6c 90 ce 42 51
0060 fc a8 ac 62 d6 fb 81 e8 12 a2 fa c5 f9
0070 85 3d 90 00 d5 c8 3d 79 80 91 c1 d8 ed
0080 b3 d4 43 f6 a2 ff 82 ff 76 4a 53 d0 a8
0090 9a 6a 05 87 5b cb 78 bd 9f ae 00 a6 62
"icm" is neither…a protocol name.
Packets: 491 · Displayed: 491 (100.0%) · Dropped: 0 (0.0%)
Profile: Default

COMPLETION QUESTIONS

1. What is the difference between the OPNsense WAN and LAN interfaces?

The WAN interface connects OPNsense to the Internet, while the LAN interface connects devices on the local network.

2. Why must both internal adapters use the same VirtualBox network name?

Both adapters must use the same network name so they can communicate with each other on the same virtual network.

3. What information does the default route provide to Ubuntu?

The default route tells Ubuntu where to send traffic that is going to other networks or the Internet.

4. Which packet exchange allows Ubuntu to learn the firewall MAC address?

The ARP request and ARP reply allow Ubuntu to learn the firewall's MAC address.

5. Why does a successful ping to 1.1.1.1 not automatically prove that DNS is working?

Because pinging 1.1.1.1 only shows that the Internet connection is working. DNS may still be broken and unable to translate website names into IP addresses.

COMPREHENSIVE SUMMARY

This lab was about connecting two virtual machines using VirtualBox, OPNsense Firewall, and Ubuntu Linux. The goal was to make sure the Ubuntu computer could communicate with the firewall, access the internet, and resolve domain names successfully.

WHAT WAS DONE

Configured the **OPNsense firewall VM** with:

WAN interface connected to **NAT**.

LAN interface connected to **ICDFA-LAN**.

Configured the **Ubuntu VM** to use the same **ICDFA-LAN** network.

Verified that Ubuntu received a valid **10.10.10.x** IP address from the firewall through DHCP.

Tested connectivity by:

Pinging the firewall (**10.10.10.1**).

Pinging an internet IP address (**1.1.1.1**).

Resolving a domain name (**opnsense.org**).

Used **Wireshark** to capture and analyse:

ARP packets (used to find MAC addresses).

ICMP packets (used for ping communication).

DNS packets (used for domain name resolution).

WHAT WAS LEARNED

The **WAN interface** connects OPNsense to the internet.

The **LAN interface** connects local devices to the firewall.

The **default gateway (10.10.10.1)** allows Ubuntu to reach other networks and the internet.

ARP helps devices discover MAC addresses.

ICMP is used by the ping command.

DNS converts domain names into IP addresses.

CONCLUSION

The lab was successful because Ubuntu received a valid IP address, communicated with the firewall, accessed the internet, resolved domain names, and captured ARP, ICMP, and DNS packets in Wireshark. This demonstrated successful network commissioning and connectivity verification between the two virtual machines.